ASSIGNMENT-2

REPORT

1. INTRODUCTION

Handwritten digit recognition is a fundamental problem in pattern recognition and machine learning. The MNIST dataset is widely used to evaluate classification algorithms. In this assignment, we build a classification pipeline using only three-digit classes: 0, 1, and 2.
The main goal of this experiment is to study how dimensionality reduction techniques affect classification performance. Three main techniques are used:

- Principal Component Analysis (PCA) for dimensionality reduction.
- Fisher's Discriminant Analysis (FDA) for maximizing class separability.
- Discriminant Analysis (LDA and QDA) for classification.

The dataset is first pre-processed and converted into feature vectors. PCA is then used to reduce dimensionality while preserving variance. FDA is applied to obtain an optimal projection that separates the classes. Finally, LDA and QDA classifiers are trained and evaluated.

2. DATASET

The MNIST dataset was downloaded from Kaggle in IDX binary format (as given in the assignment.). Each image was converted into a 784-dimensional feature vector by flattening the pixel matrix.

For this assignment, only digits 0, 1, and 2 were selected. From each class:
• 100 samples were randomly selected from the training set
• 100 samples were randomly selected from the test set

This resulted in 300 training samples and 300 test samples.

3. PCA (Principal Component Analysis)

Step-1: Mean Computation

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Step-2: Mean Centering

$$X_c = X - \mu$$

Step-3: Finding Covariance Matrix

$$S = \frac{X_c X_c^T}{N-1}$$

Or Divided by N (biased)

Step-4: Obtain eigenvectors and eigenvalues. Sort the eigenvectors in descending order of eigenvalues.

Step-5: Dimensionality Reduction

To retain **75% of total variance**, the first $p$eigenvectors are selected to form the projection matrix, Up.
The reduced representation is obtained as

$$Y = U_p^T X_c$$

Step-6: Reconstruction
The original image can be approximately reconstructed using

$$\hat{X} = U_p Y + \mu$$

The reconstruction quality is measured using **Mean Squared Error (MSE)**.

$$MSE = \frac{1}{n}\sum(x_i - \hat{x}_i)^2$$

4. <u>FDA (Fischer Discriminant Analysis)</u>

(i) Within Class Scatter

$$S_W = \sum_c \sum_{x_i \in C_c} (x_i - \mu_c)(x_i - \mu_c)^T$$

(ii) Between Class Scatter

$$S_B = \sum_c N_c(\mu_c - \mu)(\mu_c - \mu)^T$$

The optimal projection matrix W is obtained by maximizing

$$\frac{\boldsymbol{det(W^T S_B W)}}{\boldsymbol{det(W^T S_W W)}}$$

This leads to solving the generalized eigenvalue problem

$$\boldsymbol{S_B w = \lambda S_W w}$$

5. Linear Discriminant Analysis (LDA)

The discriminant function is

$$g_i(x) = x^T \Sigma^{-1} \mu_i - \frac{1}{2} \mu_i^T \Sigma^{-1} \mu_i + \ln P(\omega_i)$$

6. Quadratic Discriminant Analysis (QDA)

The discriminant function becomes quadratic:

$$g_i(x) = -\frac{1}{2} \ln |\Sigma_i| - \frac{1}{2} (x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i) + \ln P(\omega_i)$$

7. ACCURACY METRIC

Accuracy is a commonly used metric in classification tasks, measuring the proportion of correct predictions over the total number of predictions.

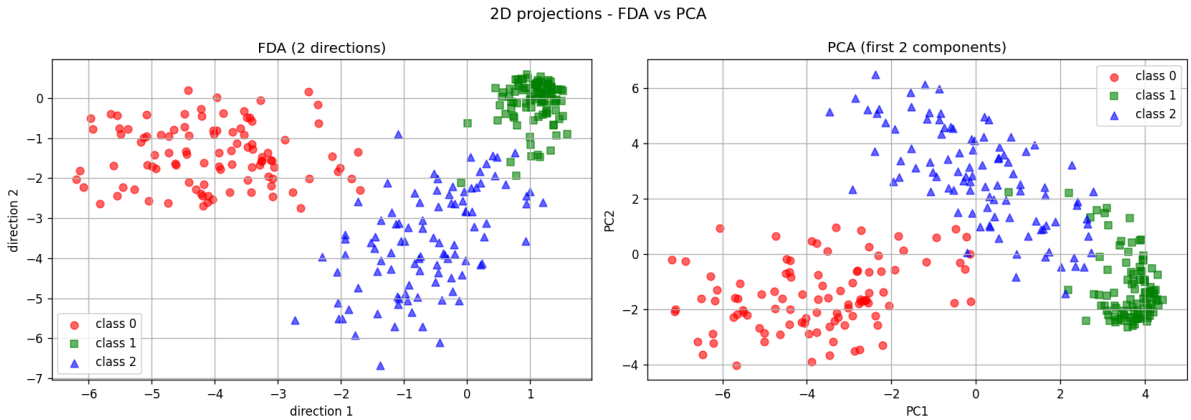
$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(\hat{y}_i = y_i)$$

8. RESULTS

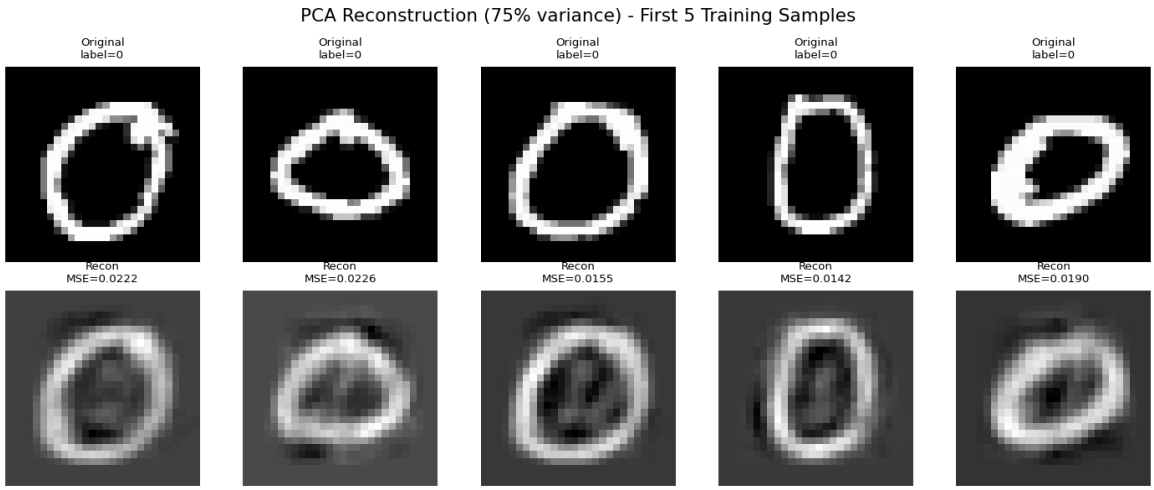
Method	Train Acc	Test Acc
FDA+LDA	96.33%	92.00%
FDA+QDA	97.67%	94.00%
PCA75%+LDA	97.67%	94.33%
PCA90%+LDA	99.00%	95.00%
PCA-2PC+LDA	93.00%	90.00%

9. Visualizations

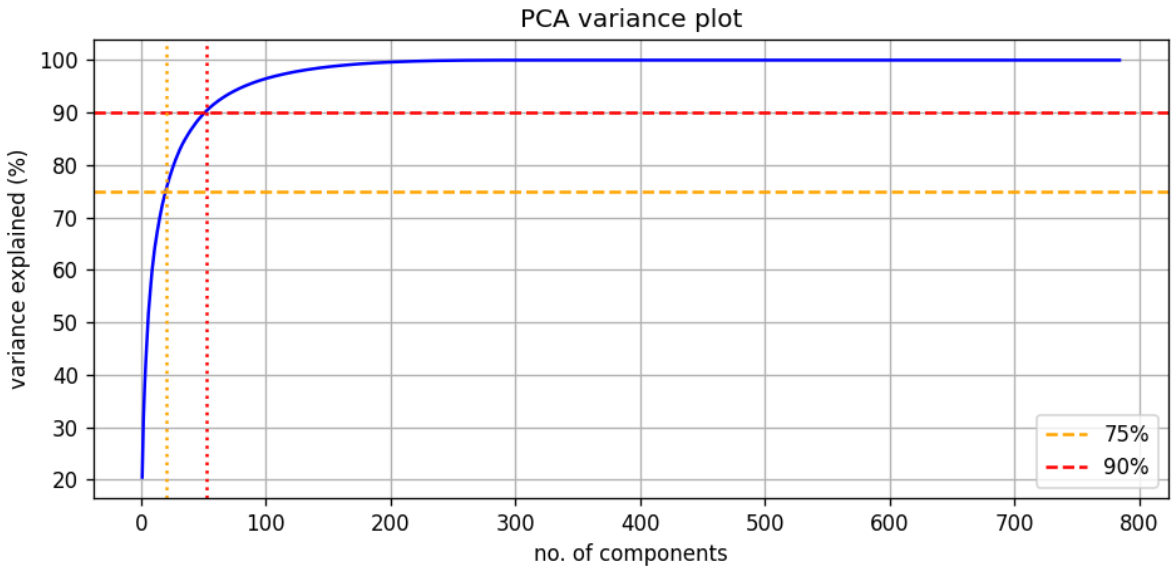
(i) 2D Projection



(ii) PCA Reconstruction



(iii) PCA Variance Plot



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10. Conclusion

Principal Component Analysis reduces the dimensionality of the data while preserving most of the variance.
In this experiment, three PCA settings were tested:

1. PCA retaining 75% variance
2. PCA retaining 90% variance
3. Using only the first two principal components

11. Observations

- When 75% variance is retained, the dimensionality is reduced significantly while still maintaining good classification performance.
- Increasing the retained variance to 90% slightly improves accuracy because more information about the original data is preserved.
- When only two principal components are used, accuracy decreases because a large amount of information is lost during dimensionality reduction.
  These results show that PCA provides a trade-off between dimensionality reduction and classification performance.